

Manh Khieu Vu

Senior Software Engineer

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ABOUT

Senior Software Engineer with 7+ years designing and shipping production C/C++ systems on Linux - spanning multi-source database engines, enterprise printer platforms, and real-time security software. I enjoy solving complex system-level problems where performance and reliability are non-negotiable, and have spent 6+ years collaborating directly with Japanese engineering teams across two companies, delivering under strict timelines and high-pressure release cycles. Growing toward Solution Architecture - where I can combine strong systems knowledge with the communication skills built from years of cross-cultural collaboration.

EXPERIENCE

Senior Software Engineer

Nov 2024 - Present

FPT Software

KMI multi-function printer · C++ · Japanese enterprise clients

Print Process Team

- Develop and maintain print workflow modules within the KMI platform - handling job reception, scheduling, and communication with hardware for printing and finishing (stapler, sorter, group)
- Implement and adapt features across multiple printer models, ensuring compatibility across hardware configurations and product variants
- Optimize job processing time, resource utilization, and system responsiveness while new features and finisher integrations are continuously introduced
- Investigate and resolve complex system-level issues including concurrency problems, resource constraints, and cross-component interactions
- Collaborate with cross-functional teams (firmware, driver, API, GUI, validation, QA) to deliver features, evaluate cross-model impact, and maintain release quality

Market Issue Team

- Manage end-user bug reports and investigation requests from the market - owning the full cycle from root cause analysis through solution, release, and closure
- Coordinate between internal modules and Japanese clients to communicate findings, assess impact, and align on corrective actions

Software Engineer

Aug 2019 - Jul 2024

TOSHIBA Software Development (Vietnam)

PGSpider - Federated Query Engine github.com/pgspider

Database Development · C · Open Source (180+ stars)

- Collaborated directly with Japanese clients to receive, clarify, and analyze software feature requirements
- Took ownership of full-feature development - design, implementation, testing, and documentation - meeting strict delivery timelines
- Contributed to PGSpider, an open-source PostgreSQL extension for federated querying across multiple heterogeneous databases
- Implemented and maintained core features including Foreign Data Wrappers (FDW) and performance enhancements for distributed SQL execution
- Integrated multiple data sources (MySQL, Oracle, MongoDB, SQLite, InfluxDB, GridDB, S3/Parquet) via FDWs, ensuring compatibility and performance
- Implemented query pushdown optimization - delegating WHERE, aggregations, ORDER BY, and GROUP BY to child nodes

Embedded Linux Multimedia Platform

Toradex Apalis iMX · Qt · GStreamer · Wayland · DBus

- Developed a multi-process multimedia system on Toradex Apalis iMX to display video, images, and text on embedded Linux
- Scaled output from Full HD to 4K resolution and from 30 fps to 60 fps through rendering and pipeline optimization
- Built media processing pipelines with GStreamer for video capture, transcoding, and playback
- Implemented Wayland/Weston compositor integration for hardware-accelerated graphics rendering
- Designed inter-process communication architecture using DBus for coordinating multiple system services
- Optimized startup time and memory footprint for resource-constrained embedded targets

Software Engineer - Security & Malware Prevention

Mar 2017 - Jul 2019

Bkav Corporation

Bkav Pro · C/C++, Python, Assembly · Windows & Linux

- Researched and developed malware prevention systems, contributing to Bkav Pro product
- Performed malware and behavioral analysis - reverse engineering malicious binaries using IDA Pro
- Built and maintained custom sandbox environments to safely execute and monitor suspicious samples
- Wrote tools and scripts in C/C++, Python, and Assembly for security products and test processes
- Integrated detection modules into Bkav's security products across Windows and Linux platforms
- Provided technical documentation and analysis reports used for product updates and internal R&D

TECHNICAL SKILLS

Languages

C/C++ (7+ yrs) Python Golang C# Java

Systems & Platforms

Linux System Programming Qt Framework GStreamer Wayland/Weston DBus Printer Platforms

Database & Data

PostgreSQL Internals FDW Development Query Optimization MySQL SQLite InfluxDB

DevOps & Build

Git Docker CMake CI/CD

Project Management

Jira Redmine

AI & Emerging

LLM Integration OpenAI APIs AI Agent Workflows

PROJECTS

Side projects and experiments available on github.com/khieuvuvm

EDUCATION

Hanoi University of Science and Technology

2015 - 2019

Bachelor of Software Engineering

GPA: 3.1/4.0

LANGUAGES

Vietnamese (Native) · English (Business Proficiency)